



Advanced Linear Algebra: Foundations to Frontiers

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0.2.4.1 Installing the BLAS ¶

The Basic Linear Algebra Subprograms (BLAS) are an interface to fundamental linear algebra operations. The idea is that if we write our software in terms of calls to these routines and vendors optimize an implementation of the BLAS, then our software can be easily ported to different computer architectures while achieving reasonable performance.

A popular and high-performing open source implementation of the BLAS is provided by our BLAS-like Library Instantiation Software (BLIS). The following steps will install BLIS if you are using the Linux OS (on a Mac, there may be a few more steps, which are discussed later in this unit.)

- Visit the <https://github.com/flame/blis>.
- Click on

Clone or download ▾

and copy <https://github.com/flame/blis.git>.

- In a terminal session, in your home directory, enter

```
git clone https://github.com/flame/blis.git
```

(to make sure you get the address right, you will want to paste the address you copied in the last step.)

- Change directory to blis:

```
cd blis
```

- Indicate a specific version of BLIS so that we all are using the same release:

```
git checkout pfhp
```

- Configure, build, and install with OpenMP turned on.

```
./configure -p ~/blis auto
make -j8
make check -j8
make install
```

The `-p ~/blis` installs the library in the subdirectory `~/blis` of your home directory, which is where the various exercises in the course expect it to reside.

- If you run into a problem while installing BLIS, you may want to consult <https://github.com/flame/blis/blob/master/docs/BuildSystem.md>.

On Mac OS-X

- You may need to install Homebrew, a program that helps you install various software on you mac. Warning: you may need "root" privileges to do so.

```
$ /usr/bin/ruby -e "$(curl -fsSL https://raw.githubusercontent.com/Homebrew/homebrew-core/master/install.rb)"
```

Keep an eye on the output to see if the "Command Line Tools" get installed. This may not be installed if you already have Xcode Command line tools installed. If this happens, post in the "Discussion" for this unit, and see if someone can help you out.

- Use Homebrew to install the gcc compiler:

```
$ brew install gcc
```

Check if gcc installation overrides clang:

```
$ which gcc
```

The output should be `/usr/local/bin`. If it isn't, you may want to add `/usr/local/bin` to "the path." I did so by inserting

```
export PATH="/usr/local/bin:$PATH"
```

into the file `.bash_profile` in my home directory. (Notice the "period" before "bash_profile")

- Now you can go back to the beginning of this unit, and follow the instructions to install BLIS.